

# Ziqi Ang — Contribution Statement

**Primarily responsible for:** Prepare dataset, trend & seasonal analysis, ( `code/03_eda.R` )

- data glimpse, calculate aggregate statistics, correlation, covariance; evaluate skewness, kurtosis
- visualize the yearly overall trends (static & interactive plots), spot anomaly, seasonal patterns
- explore initial hypothesis with quantile regression

**Collaborative tasks:** Project ideation, research question brainstorming, data quality

## Reference

- ggplot2 (<https://ggplot2.tidyverse.org/>)
- tidyverse (<https://style.tidyverse.org/syntax.html>)
- correlation (<https://corr.tidymodels.org/reference/index.html>)
- setdiff (<https://de.mathworks.com/help/matlab/ref/double.setdiff.html>)
- Histograms and frequency polygons ([https://ggplot2.tidyverse.org/reference/geom\\_histogram.html](https://ggplot2.tidyverse.org/reference/geom_histogram.html))
- R quantile regression (<https://www.r-bloggers.com/2019/01/quantile-regression-in-r-2/>)
- R moments package (<https://cran.r-project.org/web/packages/moments/moments.pdf>)

## AI Tool Disclosure

I used the following AI tools in this project:

| Tool                           | Purpose                           | Scope                                                                                                              |
|--------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------|
| ChatGPT-5<br>(free license)    | Debug, generate<br>plots template | Fix version control issue, read error messages; plots stylistic support<br>(parameters were adjusted and reviewed) |
| Claude,<br>Sonnet 5,<br>Medium | Code refactoring                  | Reduce repetitive lines, improve code quality, suggest lines                                                       |

*“I hereby confirm that I have written the part of this work myself. During the project, I mainly used ChatGPT-5 for debugging, generating the interactive plot template according to my specifications; Claude Sonnet 5 for improving code lines, reducing repetitiveness, suggesting code lines. Those suggestions were not adopted blindly but manually reviewed, adjusted, documented and tested.”*